Examiner only

Answer **all** questions.

1. (a) (i) State why sodium is stored in oil in the laboratory. [1]

....

(ii) Describe the change in appearance when a piece of freshly cut sodium is left for a few minutes. [1]

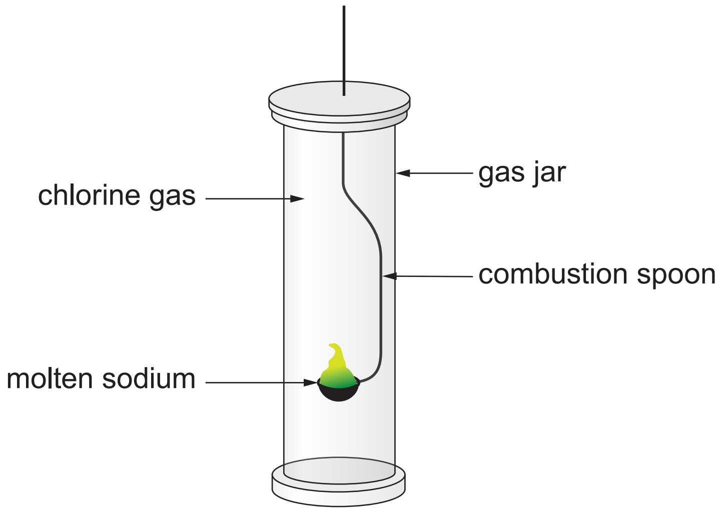
....

....

(iii) Give the formula of the compound formed when sodium reacts with oxygen. [1]

....

(b) The diagram shows the reaction of sodium with chlorine.



(i) State why it is necessary to carry out this reaction in a fume cupboard. [1]

....

(ii) Complete and balance the equation for the reaction of sodium and chlorine. [2]

$\square$ Na + Cl_2 $\longrightarrow$ $\square$ ....

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(c) The table shows some properties of Group 7 elements.

Element	Melting point (°C)	Boiling point (°C)	Reaction with hot iron
fluorine	–220	–188	explosive
chlorine	–101	–34	very fast
bromine	–7	59	quite fast
iodine	114		slow

(i) Put a tick (✓) in the box next to the most likely boiling point for iodine. [1]

–25 °C ☐ 25 °C ☐ 100 °C ☐ 150 °C ☐

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(ii) Astatine lies below iodine in Group 7. State how you would expect astatine to react with hot iron.

Give a reason for your answer. [2]

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